

Lesson 4 · Homework

NCEA Level 2 · AS 91262 · Anti-differentiation, rates and kinematics

Time: about 60 minutes

Name: _____

Date: _____

Show all working, including $+c$. Give units and meaning for every rate. [A] Achieved, [M] Merit, [E] Excellence.

1. Anti-differentiate. [A]

(a) $f'(x) = 8x^3$

.....

(b) $f'(x) = 3x^2 - 10x + 1$

.....

(c) $\frac{dy}{dx} = 5 - 2x$

.....

(d) $f'(x) = x^2 + 4x$

.....

2. $f'(x) = 2x + 3$ and $f(0) = -1$. Find $f(x)$. [A]

.....

.....

3. $f'(x) = 3x^2 - 4$ and the curve passes through $(2, 5)$. Find $f(x)$. [A]

.....

.....

4. $\frac{dy}{dx} = 6x^2 + 2x$ and the curve passes through $(-1, 3)$. Find y in terms of x . [A]

.....

.....

5. $f'(x) = 4x - 1$ and the curve passes through $(1, 2)$. Find $f(3)$. [A]

.....

.....

6. A company's revenue from selling x items is $R = 50x - 0.2x^2$ dollars. Find the rate of change of R when $x = 100$, and explain what it means. [A]

.....

.....

7. The temperature of an oven is $T = 20 + 15t - 0.25t^2$ degrees Celsius after t minutes ($0 \leq t \leq 30$). Find the rate of change at $t = 10$ and explain it. [A]

.....

.....

8. A particle moves with $s = t^3 - 9t^2 + 24t$ metres after t seconds. [A]

(a) Find its velocity at $t = 1$.

.....

(b) Find the times when it is at rest.

.....

(c) Find its acceleration at $t = 4$.

.....

9. A car has velocity $v = 3t^2 + 2 \text{ m s}^{-1}$ and starts at the origin. How far has it travelled after 4 seconds? [A]

.....

.....

10. A particle starts from rest at the origin with acceleration $a = 6 - 2t \text{ m s}^{-2}$. Find its velocity and displacement at $t = 3$. [A]

.....

.....

11. The height of a young tree is $H = 0.5 + 0.8t - 0.02t^2$ metres after t years ($0 \leq t \leq 20$). When is it growing at 0.4 m per year? [A]

.....

.....

12. A ball is thrown vertically up from a 2 m platform at 12.6 m s^{-1} (acceleration -9.8 m s^{-2}). Find its maximum height above the ground. [M]

.....

.....

.....

13. A car brakes with velocity $v = 20 - 4t \text{ m s}^{-1}$ until it stops. How far does it travel while braking? [M]

.....

.....

14. **Extension** A particle moves with $v = 3t^2 - 12t + 9 \text{ m s}^{-1}$ and starts at the origin. Find its displacement after 3 seconds, and the total distance it travels in those 3 seconds. [E]

.....

.....

.....

.....

Optional extra: StudyTime Level 2 Calculus Guide, “Quick Questions” on page 29.